Example 10.7

Differentiate the following with respect to x :

(i) $y = x^3 + 5x^2 + 3x + 7$ (ii) $y = e^x + \sin x + 2$

(iii) $y = 4\text{cosec}\, x - \log x - 2e^x$ (iv) $y = \left(x - \frac{1}{x}\right)^2$ (v) $y = xe^x \log x$

(vi) $y = \frac{\cos x}{x^3}$ (vii) $y = \frac{\log x}{e^x}$

(viii) Find $f'(3)$ and $f'(5)$ if $f(x) = |x - 4|$.

Solution

(i) $\frac{dy}{dx} = 3x^2 + 10x + 3.$ (ii) $\frac{dy}{dx} = e^x + \cos x.$

(iii) $\frac{dy}{dx} = -4\ \text{cosec}\ x.\cot x - \frac{1}{x} - 2e^x.$

(iv) $y = x^2 + \frac{1}{x^2} - 2 = x^2 + x^{-2} - 2$

$\frac{dy}{dx} = 2x - 2x^{-2-1} = 2x - \frac{2}{x^3}.$

(v) $\frac{dy}{dx} = xe^x\left(\frac{1}{x}\right) + e^x.\log x(1) + x\log x(e^x)$

$= e^x + e^x \log x + xe^x \log x = e^x(1 + \log x + x\log x).$

(vi) $y = \frac{\cos x}{x^3}$

$\frac{dy}{dx} = \frac{x^3(-\sin x) - \cos x(3x^2)}{x^6} = \frac{-x^2(x\sin x + 3\cos x)}{x^6} = -\frac{(x\sin x + 3\cos x)}{x^4}.$

(vii) $y = \frac{\log x}{e^x} = e^{-x}.\log x$

$\frac{dy}{dx} = e^{-x}\left(\frac{1}{x}\right) + \log x(e^{-x})(-1)$

$= e^{-x}\left[\frac{1}{x} - \log x\right].$

(viii) $f(x) = |x - 4| = \begin{cases} -(x-4) & ; \ x < 4 \\ (x-4) & ; \ x \geq 4 \end{cases}$

$f'(x) = \begin{cases} -1 \text{ if } & x < 4 \\ +1 \text{ if } & x \geq 4 \end{cases}$

Therefore, $f'(3) = -1$

$f'(5) = 1$.

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